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ECEN/CSEN 123 Prelab 8

Preparation:

- Study the documents related to the line follower: **QTRX-MD-08RC Reflectance Sensor Array (digital output). Make sure you have the right one!** A user's guide in the datasheet directory on google drive. A software library guide is there as well.
- The best way to install the library is to:
  - In the Arduino IDE, open the "Tools" menu, select "Manage Libraries...".
  - Search for "QTRSensors".
  - Click the QTRSensors entry in the list.
  - Click "Install".
- Determine how you will mount the sensors to your platform for this lab and obtain the necessary materials. The stock screws, nuts, and standoffs will be available in lab (#2 and #4 size) as well as some velcro.
- The QTRX-MD-08RC works best at a specific range of distances from the ground. Determine how far your sensor needs to be from your robot platform and where you will mount it.

Optimal sensing: 10mm
Max recommended: 40mm

- Design some test patterns – 1" wide black lines on white paper. Print them out or bring paper to lab and use black masking tape. You should be able to follow a curve with any radius 200mm or larger, a straight line, and up to a 45° angle.

DONE

- Print out or copy a ruler on paper that you can glue/tape to your robot. It should measure 100mm to the left and right of a center point.

DONE

- Review the line-follower library and your prior motor code. https://pololu.github.io/qtr sensors-arduino/
  Remember to include a calibration process in your system. Also note that there are several routines provided to read the sensors. You can use any or all of them. These may be most useful.

```
void read (uint16_t *sensorValues, QTRReadMode mode=QTRReadMode::On)  void
readCalibrated (uint16_t *sensorValues, QTRReadMode mode=QTRReadMode::On) uint16_t
readLineBlack (uint16_t *sensorValues, QTRReadMode mode=QTRReadMode::On)
```

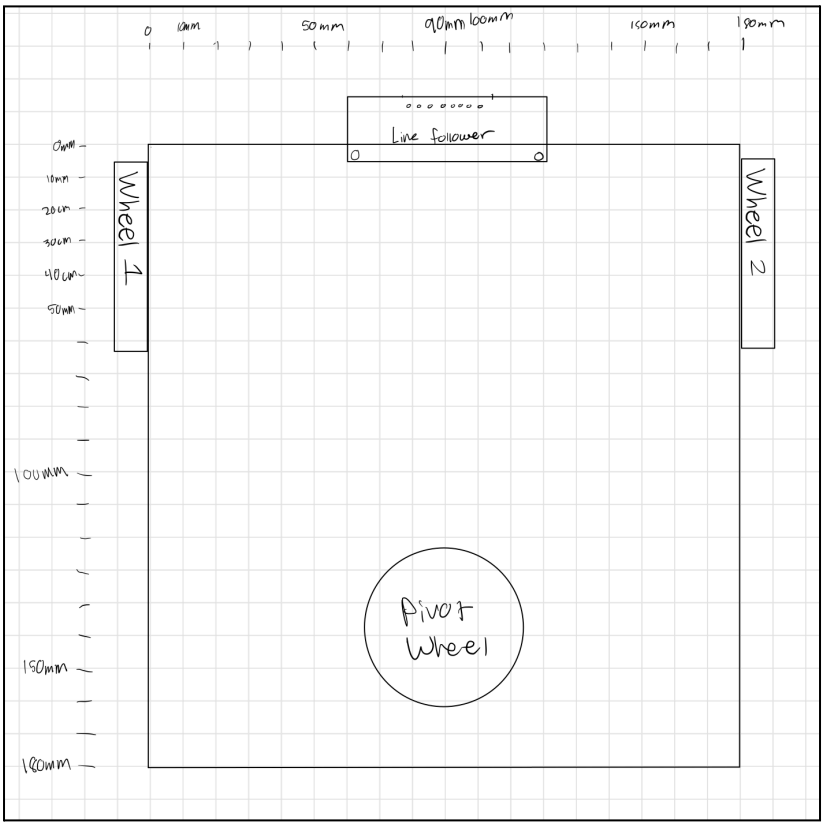
- Think about and discuss an appropriate control algorithm and how to calibrate the control parameters.

If we get some feedback from the line sensor, we could then somehow tie motor drive to that feedback. If we're veering off left, put more power on the left motor. If we're veering off right, put more power on the right motor. Finding the appropriate

balance of the rate at which we correct can be done in lab and we will aim for a critically damped system.

Pre-Lab Report:

- Create an accurate measured drawing that represents the position of each of the drive wheels and each of the line-follower photosensors on your robot. You can hand draw or use CAD. A top-projection is acceptable.

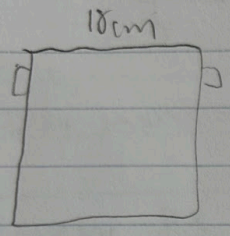


- Using these measurements, calculate how the orientation of your robot changes (i.e. the angle), when the right wheel moves forward 1cm and the left wheel simultaneously moves backwards 1cm.

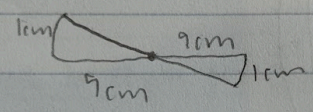
18x18

wheel diameter = 6cm

circ = 6π cm



$$\text{arc} = \frac{\theta}{360}(18\pi)$$



$$1\text{cm} = \frac{\theta}{360}(18\pi)$$

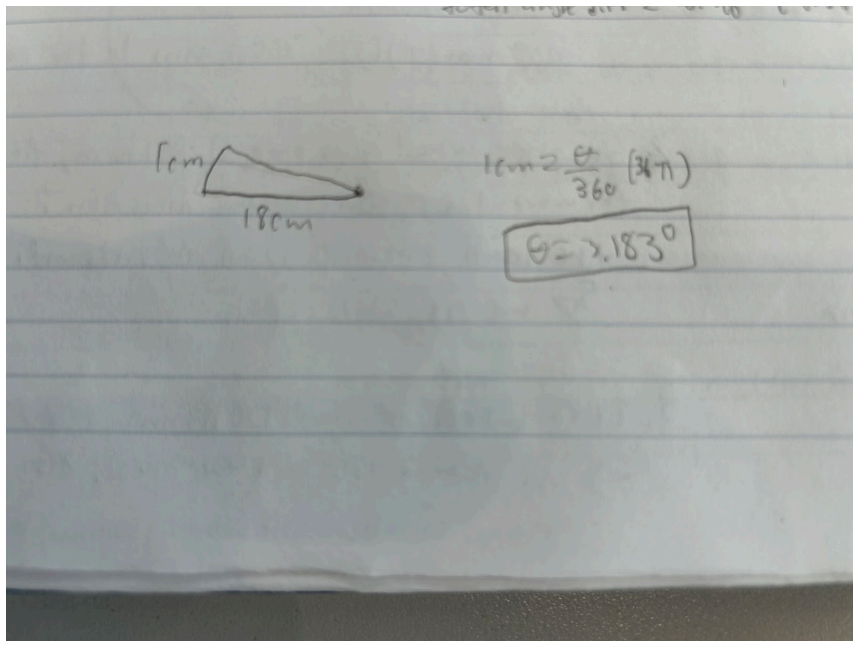
$$\theta = 6.366^\circ$$

angle of right = 6.366°

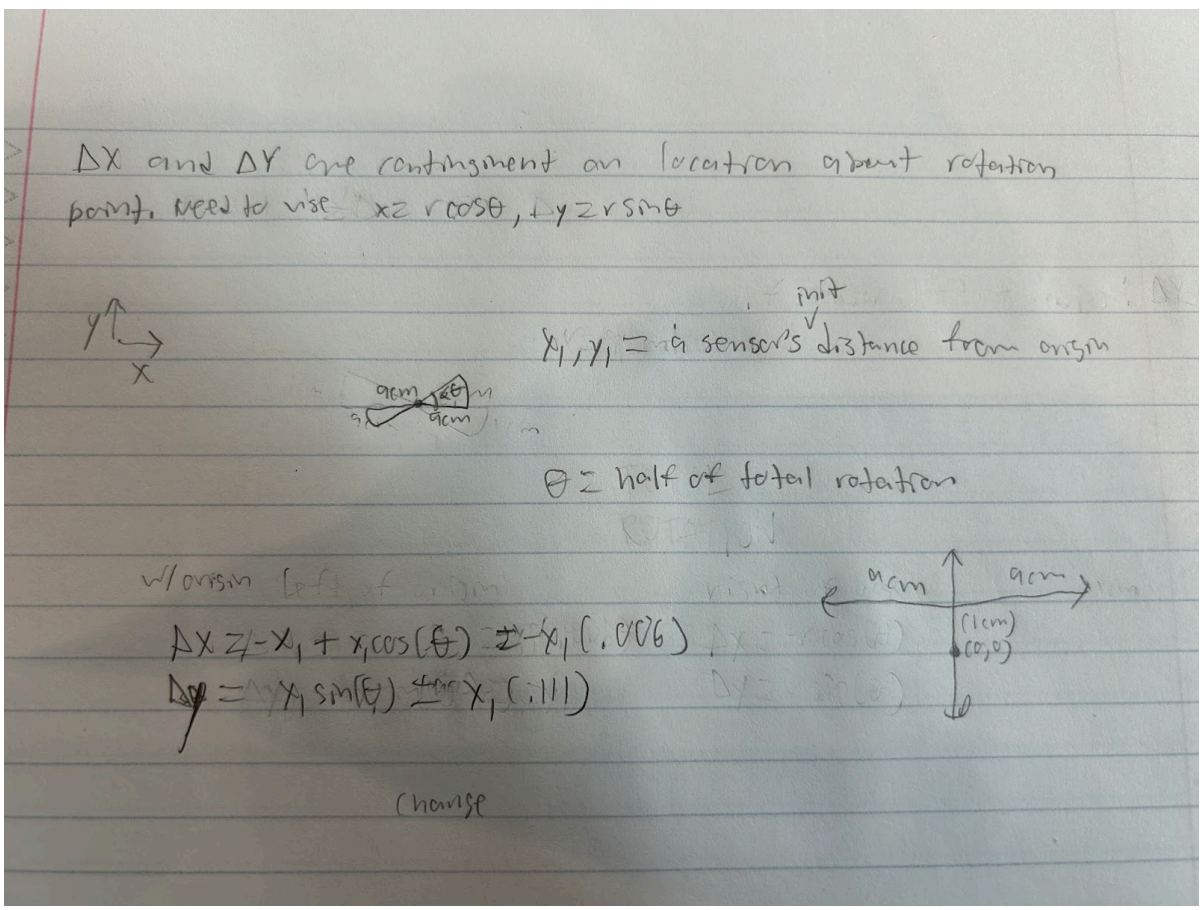
angle of left = -6.366°

total angle diff = $6.366 - (-6.366) = \boxed{12.732^\circ}$

• Using these measurements, calculate how the orientation of your robot changes (i.e. the angle), when the right wheel moves forward 1cm and the left wheel remains still.

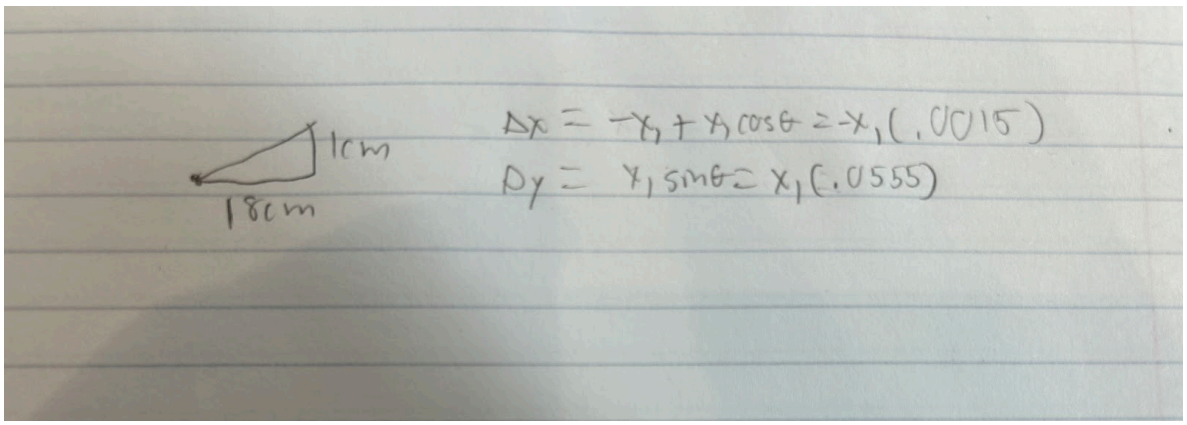


- Using these measurements, calculate how the position of each of the 8 photosensors changes (i.e. ΔX and ΔY), when the right wheel moves forward 1cm and the left wheel simultaneously moves backwards 1cm.



Sensor Number	x (cm)	y (cm)		delta x1 (cm)	delta y1 (cm)
1	-2.8	0		0.0168	-0.3108
2	-2	0		0.012	-0.222
3	-1.2	0		0.0072	-0.1332
4	-0.4	0		0.0024	-0.0444
5	0.4	0		-0.0024	0.0444
6	1.2	0		-0.0072	0.1332
7	2	0		-0.012	0.222
8	2.8	0		-0.0168	0.3108

- Using these measurements, calculate how the position of each of the 8 photosensors changes (i.e. ΔX and ΔY), when the right wheel moves forward 1cm and the left wheel remains still.



Sensor Number	x (cm)	y (cm)		delta x1 (cm)	delta y1 (cm)
1	-2.8	0		0.0042	-0.1554
2	-2	0		0.003	-0.111
3	-1.2	0		0.0018	-0.0666
4	-0.4	0		0.0006	-0.0222
5	0.4	0		-0.0006	0.0222
6	1.2	0		-0.0018	0.0666
7	2	0		-0.003	0.111
8	2.8	0		-0.0042	0.1554

- Wish you had studied harder in Trigonometry.

- Turn in all of your calculations (with explanations) and your drawings.
- Include a selfie of your team along with the social media feed of someone that each of you follows.